

HD189733b

R-2056

Benchmark run · workflow W8-benchmark-deep · record deep-bench_HD189733_260704 · created 2026-08-09T00:45:15+00:00

PRACTICE / VALIDATION RUN — not submitted to any science database

Results

Mid-Transit Time (Tmid)	2461225.799 +/- 0.0021 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.187 +/- 0.056
Transit depth (Rp/Rs)^2	3.48 +/- 2.07 [%]
Orbital Inclination (inc)	89.7 +/- 2.73
Airmass coefficient 1 (a1)	0.8 +/- 0.43
Airmass coefficient 2 (a2)	0.0 +/- 1.12
Scatter in the residuals of the lightcurve fit is	55.57 %
Best Comparison Star	None
Optimal Aperture	1.53
Optimal Annulus	13.18
Transit Duration (day)	0.086 +/- 0.013

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.187 ± 0.056 vs 0.15534 (deviation 0.56σ)
Duration fit vs published	0.086 d vs 0.0754 d (deviation 0.81σ)
Mid-time O–C	4.9 min
Depth SNR	3.3
Residual scatter	55.57 %

Light curve

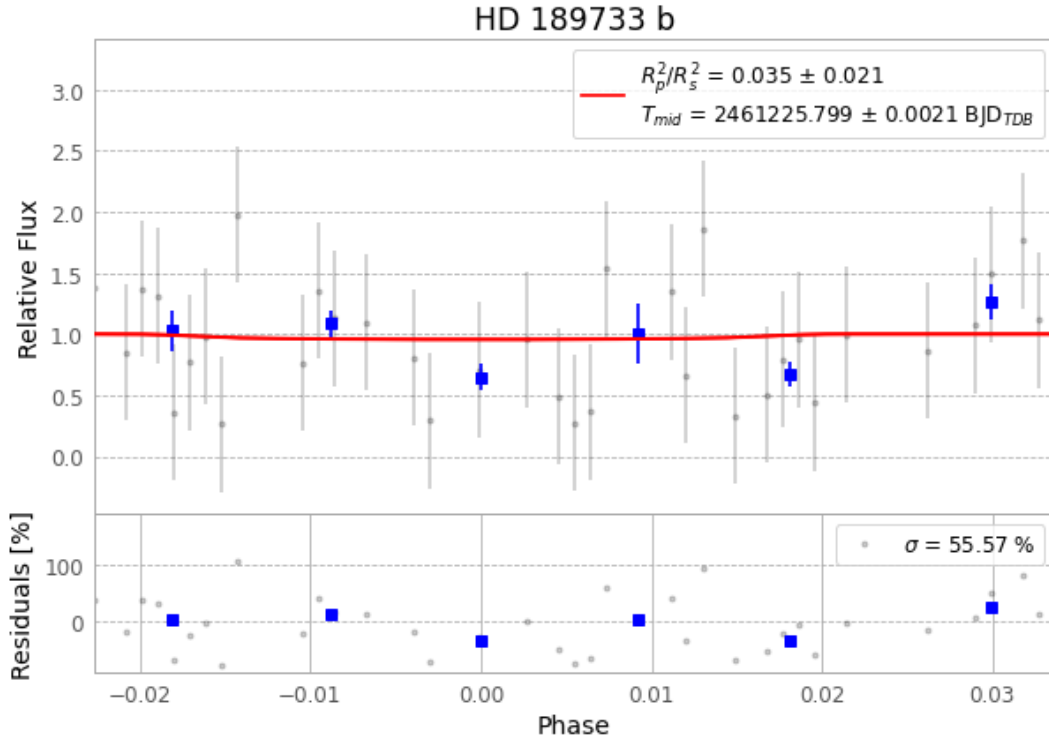


Figure 1 — FinalLightCurve_HD 189733 b_2026-07-03.png (SHA-256 b29ee4b1f35f6e99...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [215, 149], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 3d50ebd143c6552b...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit fd74a91

Data provenance

140 FITS frames (56 MB), HD189733260704050417.FITS.gz → HD189733260704085416.FITS.gz. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-HD189733-260704.log (34d2547fb620...), FinalLightCurve_HD 189733 b_2026-07-03.png (b29ee4b1f35f...), FinalLightCurve_HD 189733 b_2026-07-04.png (8450a1f5cf6f...), FinalLightCurve_HD 189733 b_2026-07-03.csv (9d52c62dea33...), FinalLightCurve_HD 189733 b_2026-07-04.csv (f91f5fc5c8db...)

Compute resources

Wall time 262.7 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)